pharmaniaga®

Cyanocobalamin 1000mcg/mL Injection

COMPOSITION:
Each 1mL ampoule contains 1000mcg Cyanocobalamin.

DESCRIPTION:
A clear and red coloured solution.

PHARMACODYNAMICS:
Cyanocobalamin is the most stable and widely used form of Vitamin B12, and has haematopoietic activity apparently identical to that of the antianaemia factor in purified liver extract. Parenteral (intramuscular) administration of Vitamin B12 completely reverses the megaloblastic anaemia and GI symptoms of Vitamin B12 deficiency. Gastrointestinal absorption of Vitamin B12 depends on the presence of sufficient intrinsic factor and calcium ions. Intrinsic factors deficiency causes pernicious anaemia, which may be associated with subacute combined degeneration of the spinal cord.

In addition, Schilling Test is a test utilising radioactive Vitamin B12 for gastrointestinal absorption of Vitamin B12 for the diagnosis of primary pernicious anaemia.

PHARMACOKINETICS:
Vitamin B12 is metabolised in the liver and has a half-life of approximately 6 days. It is excreted through the biliary system, largely unchanged in urine.

INDICATIONS:
Cyanocobalamin 1000mcg/mL Injection is used:

- For prophylaxis of anaemia associated with Vitamin B12 deficiency
- For uncomplicated pernicious anaemia or Vitamin B12 malabsorption
- For Schilling Test as a diagnostic aid to investigate Vitamin B12 malabsorption

RECOMMENDED DOSAGE:

i. Prophylaxis of anaemia
   - 250-1000mcg intramuscularly every month

ii. Uncomplicated pernicious anaemia or Vitamin B12 malabsorption
   - Initial 100 mcg daily for 5-10 days followed by 100-200 mcg monthly until complete remission is achieved. Maintenance: up to 1000 mcg monthly.
   - For children, 30-50 mcg daily for 2 or more weeks (to a total dose 1-5mg) or as prescribed.

iii. Schilling Test
   - The loading dose for Schilling Test is 1000mcg (1mg).

ROUTE OF ADMINISTRATION:
Parenteral
(Intramuscular Injection)

CONTRAINDICATIONS:
Cyanocobalamin should not be used in patient with:

i. Sensitivity to cobalt/Vitamin B12

WARNINGS AND PRECAUTIONS:
A sensitivity history should be obtained from the patient prior to administration of Vitamin B12; an intradermal test dose is recommended before Vitamin B12 is administered for the management of vitamin deficiency to patients who may be sensitive to cobalamins.

Serum potassium concentrations should be monitored during early Vitamin B12 therapy and potassium administered if necessary, since fatal hypokalemia could occur upon conversion of megaloblastic anemia to normal erythropoiesis with Vitamin B12 as a result of increased erythrocyte potassium requirements. Because Vitamin B12 deficiency may suppress the signs of polycythemia vera, treatment with cyanocobalamin may unmask this condition. The increase in nucleic acid degradation produced by administering Vitamin B12 to Vitamin B12-deficient patients could result in gout in susceptible individuals. Therapeutic response to Vitamin B12 may be impaired by concurrent infection, uremia, folic acid or iron deficiency, or by drugs having bone marrow suppressant effects (e.g., chloramphenicol).

Folic acid should be administered with extreme caution to patients with undiagnosed anaemia, since folic acid may obscure the diagnosis of pernicious anaemia by alleviating hematologic manifestations of the disease while allowing neurologic complications to progress. This may result in severe nervous system damage before the correct diagnosis is made. Vitamin preparations containing folic acid should be avoided by patients with pernicious anaemia because folic acid may actually potentiate neurologic complications of Vitamin B12 deficiency. Conversely, doses of cyanocobalamin exceeding 10 mcg daily may improve folate-deficient megaloblastic anaemia and obscure the true diagnosis.

Cyanocobalamin should not be used in patients with early Leber's disease (hereditary optic nerve atrophy), since rapid optic nerve atrophy has been reported following administration of the drug to these patients. Vitamin B12 is contraindicated in patients who have experienced hypersensitivity reactions to the vitamin or to cobalt.

INTERACTIONS WITH OTHER MEDICAMENTS
Serum concentrations of cyanocobalamin may be lowered by oral contraceptives but this interaction is unlikely to be significance but should be taken into account when performing assays for blood concentrations. Parenteral chloramphenicol may attenuate the effect of Vitamin B12 in anaemia.

Incompatibilities
Vitamin B12 injection has been reported to be incompatible with chlorpromazine hydrochloride, phytonadione, prochlorperazine edisylate, warfarin sodium, ascorbic acid, dextrose, heavy metals, oxidizing or reducing agents, and alkaline or strongly acidic solutions.

STATEMENT ON USAGE DURING PREGNANCY AND LACTATION
Pregnancy
Vitamin B12 requirements are increased in pregnant women. Parenteral preparations should be used during pregnancy only when the potential benefits justify the potential risks to the fetus.
Lactation
Vitamin B12 is distributed into human milk. Therefore, it is not recommended for breastfeeding mothers unless the expected benefits to the mother outweigh the potential risk to the infant.

ADVERSE EFFECTS/UNDESIRABLE EFFECTS
Allergic hypersensitivity reactions have occurred rarely after cyanocobalamin and include skin rashes such as rash and itching, and anaphylaxis. Patients who are hypersensitive to cyanocobalamin injection may be able to take oral cyanocobalamin. Injection site reactions including pain, erythema, pruritus, induration, swelling, and necrosis can occur.

Other adverse effect reported with cyanocobalamin include gastrointestinal disturbances, fever, chills, hot flushing, dizziness, malaise, acneform and bullous eruptions, and tremor. Cyanocobalamin should, if possible, not be given to patients with suspected Vitamin B12 deficiency without first conforming the diagnosis. Regular monitoring of the blood is advisable. Use of doses greater than 10 micrograms daily may produce haematological response in patients with folate deficiency; indiscriminate use may mask the precise diagnosis. Conversely, folate may mask Vitamin B12 deficiency.

Cyanocobalamin should not be used in Leber's disease or tobacco amblyopia since these optic neuropathies may degenerate further.

OVERDOSE AND TREATMENT
Treatment is unlikely to be needed in cases of overdosage.

STORAGE CONDITIONS
Store below 30°C. Protect from light.
Do not use if solution contains growth, particles, turbidity or if there is any change of appearance.

SHELF LIFE:
2 years from date of manufacture.

DOSAGE FORMS AND PACKAGING AVAILABLE:
10 x 1mL ampoule (amber).

PRODUCT REGISTRATION HOLDER/MANUFACTURER:
PHARMANIAGA LIFESCIENCE SDN BHD (198201002939)

Lot 7, Jalan PPU 3
Taman Perindustrian Puchong Utama
47100 Puchong, Selangor Darul Ehsan,
Malaysia

2002036-V

Revision date: 10 August 2020